❑ The Metric System

- In 1783, James Watt first proposed the use of decimal system and the commensurability of weight, length and volume. Metric weights were officially recognized in Great Britain in 1864, in United States in 1866 and in India in 1956. The principal attributes of this system are its simplicity, brevity and adaptability to everyday needs. It is the system of decimal progression meaning thereby that every unit is multiplied or divided by the same number (i.e. 10) to obtain various denominations. Latin prefixes like *milli, centi, deci,* etc. are used to indicate subdivisions while Greek prefixes like *deca, hecto, myria,* are used to denote multiples of the principal units.
- A brief account of different systems of weights and measures and relationships relevant to a pharmacist are given below.

1. Measures of Mass

- The difference between weight and mass is that the weight changes with the altitude, as it is dependent on the gravitational force, whereas mass remains unchanged.

Kilogram.

Table 1.1: Measures of mass and their Equivalents

UNITS	EQUIVALENTS
1 Tonne (T)	1,000 kilograms (Kg or kg or kilo)
1 Quintal	100 kilograms (kg)
1 kilogram (kg)	1000 grams (g, G or Gm or gm)
1 Hectogram (Hg)	100 grams
1 Dekagram (Dg)	10 grams
1 Decigram (dg)	1/10 gram = 0.1gm
1 Centigram (cg)	1/100 gram = 0.01gm
1 Milligram (mg)	1/1000 gram=0.001gm
1 Microgram (mcg or μg)	1/1000 mg or 10^{-6} gm
1 Nanogram	1/1000 μg = 10^{-9} gm
1 Picogram	1/1000 of a nanogram = 10^{-12} gm
1 Gram (g)	1000 mg

2. Measures of Length

- The unit of length is the Meter (M), which is defined as the length of the International Prototype bar

Table 1.2: Measures of Length and their Equivalents

UNITS	EQUIVALENTS
1 Kilometer (km)	1000 meters (m)
1 Hectometer (hm)	100 m
1 Dekameter (dam)	10 m
1 Decimeter (dm)	1/10 meter (0.1 m)
1 Centimeter (cm)	1/100 meter (0.01m)
1 Millimeter (mm)	1/1000 meter (0.001 m)
1 Micron (μ)	1/1000 mm (0.001 mm = 10^{-6} m)
1 Micrometer (μm)	10^{-6} m
1 Nanometer (nm)	10^{-9} m
1 Micromicron (μμ)	10^{-9} mm (10^{-12} m)

3. Measures of Capacity

- The unit of capacity is the litre or liter (l or L.) that is defined as the volume of 1 kg of water. There is difference between millilitre and cc. 1 ml is the volume occupied by 1 gm of water whereas 1 cc is the volume of a cube each side of which is 1 cm length. 1 cc ≈ 0.9998 1 litre = 1000.028 cc.
- However, for all practical purposes ml and cc are taken as the same.

Table 1.3: Measures of Capacity and their Equivalents

UNIT	EQUIVALENT
1 Kilolitre (kL)	1000 litres (L)
1 Hectolitre (hL)	100 L
1 Dekalitre (daL)	10 L
1 Litre	1000.028 cc = 1000 ml
1 Microlitre (μl)	0.001 ml = 0.000001 L

1 Millilitre	0.001 L
1 Centiliter (cL)	0.01 L
1 Deciliter (dL)	0.1 L
1000 litres	1 cubic metre (1 m^3)

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